

Implementation shortfall is then computed as the difference between paper return and portfolio return as follows:

$$IS = \underbrace{S \cdot P_n - S \cdot P_d}_{\text{Paper Return}} - \underbrace{\left(\sum s_j \right) \cdot P_n - \sum s_j p_j - fees}_{\text{Actual Portfolio Return}} \quad (3.4)$$

In our example above, the implementation shortfall for the order is:

$$IS = \$5000 - \$2400 = \$2600$$

The implementation shortfall metric is a very important portfolio manager and trader decision making metric. It is used to select stock picking ability, measure trading costs, and as we show below, measure broker and algorithmic performance.

Implementation shortfall can be described in terms of the following three examples:

1. Complete Execution
2. Opportunity Cost (Andre Perold)
3. Expanded Implementation Shortfall (Wayne Wagner)

Complete Execution

Complete execution refers to the situation where the entire order is transacted in the market. That is $\sum s_j = S$. Suppose a manager decides to purchase S shares of stock that is currently trading at P_d and at the end of the trading horizon the price is P_n . Then implementation shortfall is computed following the above calculation as follows:

$$IS = (S \cdot P_n - S \cdot P_d) - \left(\left(\sum s_j \right) \cdot P_n - \sum s_j p_j - fees \right)$$

Since $\sum s_j = S$ this equation reduces to:

$$IS = \sum s_j p_j - S \cdot P_d + fees$$

This could also be written in terms of the average execution price P_{avg} for all shares as follows:

$$IS = S \cdot P_{avg} - S \cdot P_d + fees = S \cdot (P_{avg} - P_d) + fees$$

since $\sum s_j p_j = S \cdot P_{avg}$. Notice that when all shares are executed the implementation shortfall measure does not depend on the future stock price P_n at all.